Date: _____ Day: _____

4.5 Conditional Probability.

CLASS Notes:

Condition Probability:- Prob that event B occurs given that event A occurs is called a conditional Prob. If is denoted By: P(B|A).

↳ prob of B given that A

Imp Note:- Whenever we get additional info our sample space is reduced and when prob is calcupted by reduce sample space its known as conditionay Probabi Lity.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)/n(S)}{n(B)/n(S)}$$

Class Content

Question: 37 (From Slide # 8 Class)

A math Teacher gave her class two test. 25 % of class passed both test and 42 % of class passed First test. What percent of those who passed first also past the second test.

Let event T_1 : Passed the First Test

Let event T_2 : Passed the second Test

Date: _____ Day: _____

Given:-

$P(T_1 \cap T_2) = 25\% = 0.25$

$P(T_1) = 42\% = 0.42$

Find:

$P(T_2 | T_2) = ?$

Solution:

$\therefore P(T_2 | T_1) = \frac{P(T_2 \cap T_1)}{P(T_1)} = \frac{0.25}{0.42}$

$P(T_2 | T_1) = 0.5952$

$P(T_2 | T_1) = 59.52\%$

Question:38 (From Slides & Class).

Let event W: White Marbles

Let event B: Black Marbles

Given Data:

$P(W \cap B) = 0.34$

$P(B) = 0.47$

Find:

$P(W | B) = ?$

Qs: Jar contain black & White marbles. Prob of selecting black marble and then white is 0.34. and Prob of selecting a black on First draw is 0.47. Find Prob of selecting white given First marble was drawn was black.

$\therefore P(W|B) = \frac{P(W \cap B)}{P(B)} = \frac{0.34}{0.47}$

Date: _____ Day: _____

$P(W|T) = 0.7234$

$P(W|T) = 72.34\%$

Question: 39 (From class 8 slides)

At kennedy Middle school, prob that a student takes Technology & spanish is 0.087. Prob that student takes Techn is 0.68. What is prob that a student takes spanish given that student is Taking Technology.

Given:

Let event T: Std take Technology.

Let event S : Std Take Spanish

Given:

$P(S \cap T) = 0.087$

$P(T) = 0.68$

Find:

$P(S|T) = ?$

$\therefore P(S|T) = \frac{P(S \cap T)}{P(T)} = \frac{0.087}{0.68} = 0.1279$

$P(S|T) = 12.79\%$

Date: _____ Day: _____

Question : 39 (b) (From slides & class)

A laboratory blood Test is 98% effective in detecting certain Disease if person has Disease. However the test also yeilds a "False positive" result for 0.5% of healthy person tested. (That is if healthy person is tested, then with prob 0.005 the test result will show positive). Assume that 2% of popu actually has this Diegease. What is Prob a person has disease given that test result is +ive.

Let event D: Person has Disease
Let event D': Person is healthy (no Disease)
Let Event T: Test result is positive.

Scenario 1:-
Disease exist & ALSO predicted as positive

Scenario 2:
Disease not exist & ~~ALSO pre~~ still predicted as +ive.

Date: Day:

$P(D) = 2\% = 0.02$

$P(D') = 1 - 0.02 = 0.98$

$P(T|D) = 98\% = 0.98$ → scen: 1

$P(T|D') = 0.5\% = 0.005$ → scen: 2

Baye's Theorem:

$$\therefore P(D|T) = \frac{P(T|D)\,P(D)}{P(T)}$$

$$P(D|T) = \frac{P(T|D)\,P(D)}{P(T|D)P(D) + P(T|D')P(D')}$$

$$P(D|T) = \frac{(0.98)(0.02)}{(0.98)(0.02) + (0.005)(0.98)}$$

$$P(D|T) = \frac{0.0196}{0.0196 + 0.0049}$$

$$P(D|T) = \frac{0.0196}{0.245} = 0.8$$

$P(D|T) = 80\%$

Example: 4.16

its an easy example see qns on Pg # 174.

Date: _____ Day: _____

Example: 4.17

qns on Book pg # 175

a-

Determine prob (unconditional) that selected Faculty member is in his or her 50s.

$$\therefore P(A_4) = \frac{253}{1164} \leq 0.217$$

$$\boxed{P(A_4) = 21.7\%}$$

b-

Determine prob (conditional) that selected family member is in his or her 50s given that an assistant professor is selected.

$$P(A_4 | R_3) = \frac{36}{320} = 0.1125 = \frac{P(A_4 \cap R_3)}{P(R_3)}$$

$$\boxed{P(A_4 | R_3) = 11.3\%}$$

Example: 4.19

qns on Pg # 177

a-

Prob that the adult selected is divorced given that adult selected is a male

Date: Day:

$$\therefore P(M_4 | S_1) = \frac{P(M_4 \cap S_1)}{P(S_1)}$$

$$P(M_4 | S_1) = \frac{0.044}{0.484} = 0.0909$$

$$\boxed{P(M_4 | S_1) = 9.09\%}$$

b-

Prob that adult selected is a male given that adult selected is divored.

$$\therefore P(M_4 | S_1) = \frac{P(M_4 \cap S_1)}{P(S_1)}$$

$$P(M_4 | S_1) = \frac{0.044}{0.104} = 0.423$$

$$\boxed{P(M_4 | S_1) = 42.3\%}$$

Exercise - 4.5

Question: 4.106

qs Page 178

Given Data:-

Sample Space = {HH, HT, TH, TT}

A = event fist toss is Head = {HH, HT}

B = event second toss is Head = {HH, TH}

C = event atleast one toss is Head = {HH, HT, TH}

Date: _____ Day: _____

solution:

a-

P(B):

$\therefore P(B) = \frac{2}{4} = 0.5$

$\boxed{P(B) = 50\%}$

b-

P(B|A): Prob ~~that given~~ that second toss is Head given that Fist is Head

$\therefore P(B|A) = \frac{P(B \cap A)}{P(A)}$

Reduce

Sample space = {HH, HT}

$P(B|A) = 1/2 = 0.5$

$\boxed{P(B|A) = 50\%}$

c-

P(B|C): ~~Conditional Pr~~ Prob ~~given~~ that second toss is head given that atleast one is Head)

Sample Space = {HH, HT, TH}

$P(B|C) = 2/3 = 0.667$

$\boxed{P(B|C) = 66.7\%}$

Day:

d:

P(C) :-

$P(C) = 3/4 = 0.75$

$\boxed{P(C) = 75\ \%}$

e-

P(C|A) :- Prob the atleast one is head given that First toss is Head.

SampleSpace = {HH, HT}

$P(C|A) = 3/2 = 1$

$\boxed{P(C|A) = 100\ \%}$

f-

P(C|(not B)) : Prob that atLeast one is Head given that second toss is not head

Sample Space = {HH, HT}

$P(C|(\text{not } B)) = \frac{1}{2} = 0.5$

$\boxed{P(C|\text{not } B)) = 50\ \%}$

Date: _____ Day: _____

Question: 4.107

Given Data:-

Sample Space = 52

A = event a face card is select = 12

B = event a king is selected = 4

C = event a Heard is selected = 13

a-

P(B)

$P(B) = \frac{4}{52} = 0.0769$

P(B) = 7.69%

b-

P(B|A):- Sample Space = 12

$P(B|A) = \frac{4}{12} = \frac{1}{3}$

P(B|A) = 0.333

P(B|A) = 33.3 %

c-

P(B|C)

Sample Space = 13

$P(B|C) = \frac{1}{13} = 0.077$

P(B|C) = 7.7 %

d:

$P(B \mid (\text{not } A))$

Sample space = 40

$P(B \mid (\text{not } A)) = \frac{0}{40} = 0$

$P(B \mid (\text{not } A)) = 0\ \%$

e-

$P(A)$:

Sample space = 52

$P(A) = \frac{12}{52} = 0.231$

$P(A) = 23.1\ \%$

f-

$P(A \mid B)$:-

Sample space = 4

$P(A \mid B) = \frac{4}{4} = 1$

$P(A \mid B) = 100\ \%$

g-

$P(A \mid C)$:-

Sample space = 13

$P(A \mid C) = \frac{3}{13} = 0.231$

$P(A \mid C) = 23.1\ \%$

Date: _____ Day: _____

i-

P(A | (not B)):

sample space = 48

$P(A \mid (\text{not } B)) = \frac{8}{46} = 0.167$

P(A | (not B)) = 16.7%

Question: 4.109

Qs: Page no 178

(a) Prob that the unit has exactly Four rooms

(b) Conditional Prob that the unit has exactly Four rooms, given it has at Least two rooms.

(c) Conditional Prob that unit has at most Four rooms given that it has at Least two rooms.

(d) Interpret your answers in part (a) & (c) in terms of percentage.

n = 128,202

Date: _____ Day: _____

a:

Let event A: unit has exactly Four rooms.

$$\therefore P(A) = \frac{\text{Freq of 4-room Unit}}{\text{Freq of ALL Units}} = \frac{f}{n}$$

$$P(A) = \frac{23,290}{128,202} = 0.182$$

$$\boxed{P(A) = 18.2\ \%}$$

b-

Let event B: Unit has Atleast two rooms

Let event C: Unit has exactly Four rooms

$$P(C|B) = ?$$

$$\therefore P(C|B) = \frac{P(C \cap B)}{P(B)} = \frac{23,290}{P(B)}$$

$$P(C|B) = \frac{23,290}{385 + 11050 + 23290 + 29186 + 27146 + 17631 + 17825}$$

$$P(C|B) = \frac{23290}{127513} = 0.183$$

$$\boxed{P(C|B) = 18.3\%}$$

Date: Day:

c-

Let event A: Unit has atmost Four rooms

Let event B: It has at Least two rooms.

$P(A|B) = ?$

$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1385 + 11050 + 23290}{127513}$

$P(A|B) = 0.280$

$P(A|B) = 28\ \%$

d:

For part a:

18.2% of all U.S housing units have exactly Four rooms.

For Part b:

Among U.S housing units that have at Least two rooms 18.3% have exactly Four rooms.

For Part c:

Among U.S housing unit that have at Least two rooms 28% have atmost Four rooms.

Date: Day:

Question: 4.110

QS Page 178

Don't use conditional Prob rule: Compute Directly.

a) A Black b) A White Female

c) male given that Filer was white

d) male given that Filer was black.

a:

$$P(R_2) = \frac{147}{407} = 0.361$$

$$P(R_2) = 36.1\%$$

b-

$$P(G_2 \cap R_1) = \frac{210}{407} = 0.516$$

$$P(G_2 \cap R_1) = 51.6\%$$

Date: _____ Day: _____

c-

$$P(G_1 | R_1) = \frac{P(G_1 \cap R_1)}{P(R_1)} = \frac{30}{240} = 0.125$$

$P(G_1 | R_1) = 12.5\%$

d-

$$P(G_1 | R_2) = \frac{P(G_1 \cap R_2)}{P(R_2)} = \frac{26}{147} = 0.177$$

$P(G_1 | R_2) = 17.7\%$

Question: 4.111

Qs Pg 178-179

a-

is a rookie

$$P(Y_1) = \frac{11}{65} = 0.169$$

$P(Y_1) = 16.9\%$

b-

weighs under 200 pounds.

$$P(W_1) = \frac{8}{65} = 0.123 = 12.3\%$$

Date: _____ Day: _____

c-

is a rookies given that weighs under 200 pounds.

$$P(Y_1 | W_1) = \frac{P(Y_1 \cap W_1)}{P(W_1)} = \frac{3}{8} = 0.375$$

$$P(Y_1 | W_1) = 37.5\%$$

d-

Weighs under 200 given that he is a rookies

$$P(W_1 | Y_1) = \frac{P(W_1 \cap Y_1)}{P(Y_1)} = \frac{3}{11}$$

$$P(W_1 | Y_1) = 0.273$$

$$P(W_1 | Y_1) = 27.3\%$$

e-

Interpret your answer in parts a-d in terms of percentages.

a): 16.9% of players on patriots roster are rookies

(b): 12.3% of player on patriots roster are weigh under 200 pounds.

Date: Day:

c): of those players who weigh under 200 pounds 37.5% are rookies.

d):- of those players who are rookies weigh 27.3% weigh under 200 pounds

Question: 4.113:

Qs Page 179

a:

Lives with Spouse

$P(L_2) = 0.510 - 0.510 = 51\%$

b-

is over 64

$P(A_4) = 0.152 = 15.2\%$

d-

Date: _____ Day: _____

c-

Lives with spouse and is over 64

$$P(L_2 \cap A_4) = \underline{0.083} = 0.083$$

$$P(L_2 \cap A_4) = 8.3\%$$

d-

Lives with spouse given that person is over 64

$$P(L_2 | A_4) = \frac{P(L_2 \cap A_4)}{P(A_4)} = \frac{0.083}{0.152} = 0.546$$

$$P(L_2 | A_4) = 54.6\%$$

e-

is over 64, given that person lives with spouse

$$P(A_4 | L_2) = \frac{P(A_4 \cap L_2)}{P(L_2)} = \frac{0.083}{0.510}$$

$$P(A_4 | L_2) = 0.163$$

$$P(A_4 | L_2) = 16.3\%$$

Date:_____ Day:_____

f-

Interpret your answers is part a-e in terms of % tages.

a):- 51% of US citize aged 15 or older Live with a spouse.

b):- 15.3% of citizen aged 15 or over are over 64 years old.

c):- 8.3 % of citizen aged 15 or older are over 64 and live with spouse.

d):- of those citizens who are over 64 years old 54.6% Live with spouse

e):- of those citizens who Live with Spouse 16.27% are over 64 years old.

Question: 4.115

Qs Pg 180.

a-

$P(T_1) = ?$

$P(T_1) = 0.441 - 0.441 = 44.1\%$

Date: _____ Day: _____

b-

$P(\text{not } A_2)$

$P(A_2) = 0.314 = 0.314$

use complement rule

$\therefore P(\text{not } A_2) = \text{TOTAL} - P(A_2)$

$P(\text{not } A_2) = 1.000 - 0.314$

$P(\text{not } A_2) = 0.686$

$P(\text{not } A_2) = 68.6\%$

c-

$P(A_4 \& T_5)$

$P(A_4 \cap T_5) = 0.022 = 0.022 = 2.2\%$

d-

$P(T_4 | A_1)$

$\therefore P(T_4 | A_1) = \frac{P(T_4 \cap A_1)}{P(A_1)} = \frac{0.070}{0.526}$

$P(T_4 | A_1) = 0.133$

$P(T_4 | A_1) = 13.3\%$

Date: _____ Day: _____

e-

$P(A_1 | T_4)$

$\therefore P(A_1 | T_4) = \frac{P(A_1 \cap T_4)}{P(T_4)} = \frac{0.070}{0.122} = 0.5738$

$P(A_1 | T_4) = 57.38\%$

Question: 4.117

Qs Pg 180

Data:-

Let event A:

Property crime is committed in rural Areas

Let event B:-

Property crimes are Burglary and committed in rural areas

$P(R) = 5.1\% = 0.051$

$P(B \cap R) = 1.6\% = 0.016$

Find:-

What percentage of crime committed in rural areas are bulgary?

$P(B | R) = ?$

$\therefore P(B | R) = \frac{P(B \cap R)}{P(R)} = \frac{0.016}{0.051}$

Date: Day:

$P(B|R) = 0.3137$

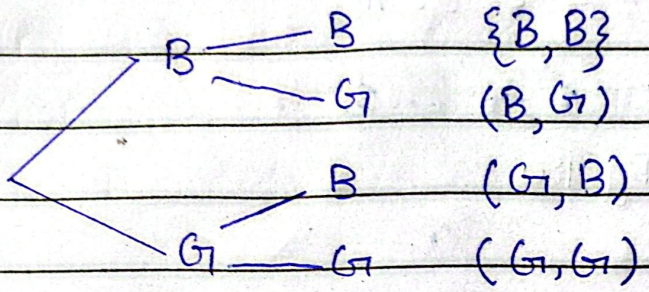
$P(B|R) = 31.37\%$

Question : 4.119

Qs Pg 180

Given Data:-

Total Children = 2



Sample space $= \{(B,B), (B,G), (G,B), (G,G)\}$

a:

Let event A: Both children are boys

Let event B: First child Born is a boy

$P(A|B) = ?$

$$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{2} = 0.5$$

$P(A|B) = 5\%$

Date: _____ Day: _____

b-

Let event A: both children are Boys

Let event C: atLeast one child is Boy

$$\therefore P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{1}{3} = 0.3333$$

$$P(A|C) = 33.33\%$$

Question: 4.121

Qs Pg 180

a:

Let A and B two events

1. **Positive Correlated**: When event A occurs, event B becomes more likely to occur.

2. **Negative correlated**: When event A occurs event B becomes less likely to occur.

3. **Independent**: The occurrence of event A does't affect the Prob. of event B.

b-

Positive correlation mean:-

$$P(A \cap B) > P(A)\,P(B)$$

Date: Day:

For B positively correlated with A:

$$P(A \cap B) > P(A)P(B)$$

Since multiplication is commutative:

$$P(A)(B) = P(B)P(A)$$

Therefore:

$$P(A \cap B) \geq P(B)P(A)$$

Above eq. means A is +ively correlated with B so Proved.

C-

Negative correlation

$$P(A \cap B) < P(A)(B)$$

For B negatively correlated with A

$$P(A \cap B) < P(A)P(B)$$

Since

$$P(A)P(B) = P(B)P(A)$$

So

$$P(A \cap B) < P(B)P(A)$$

Above eq. means A is negatively correlated with B so proved.

Date: _____ Day: _____

d.

Independent event mean:-

$$P(A \cap B) = P(A)P(B)$$

For B independent of A:-

$$P(A \cap B) = P(A)P(B)$$

Since

$$P(A)\,P(B) = P(B)P(A)$$

So:

$$P(A \cap B) = P(B)P(A)$$

↳

above eq: means A is independent of B

So Proved.